Introduction to Numerical Weather Prediction

Chapter 2: Numerical Methods

Objective

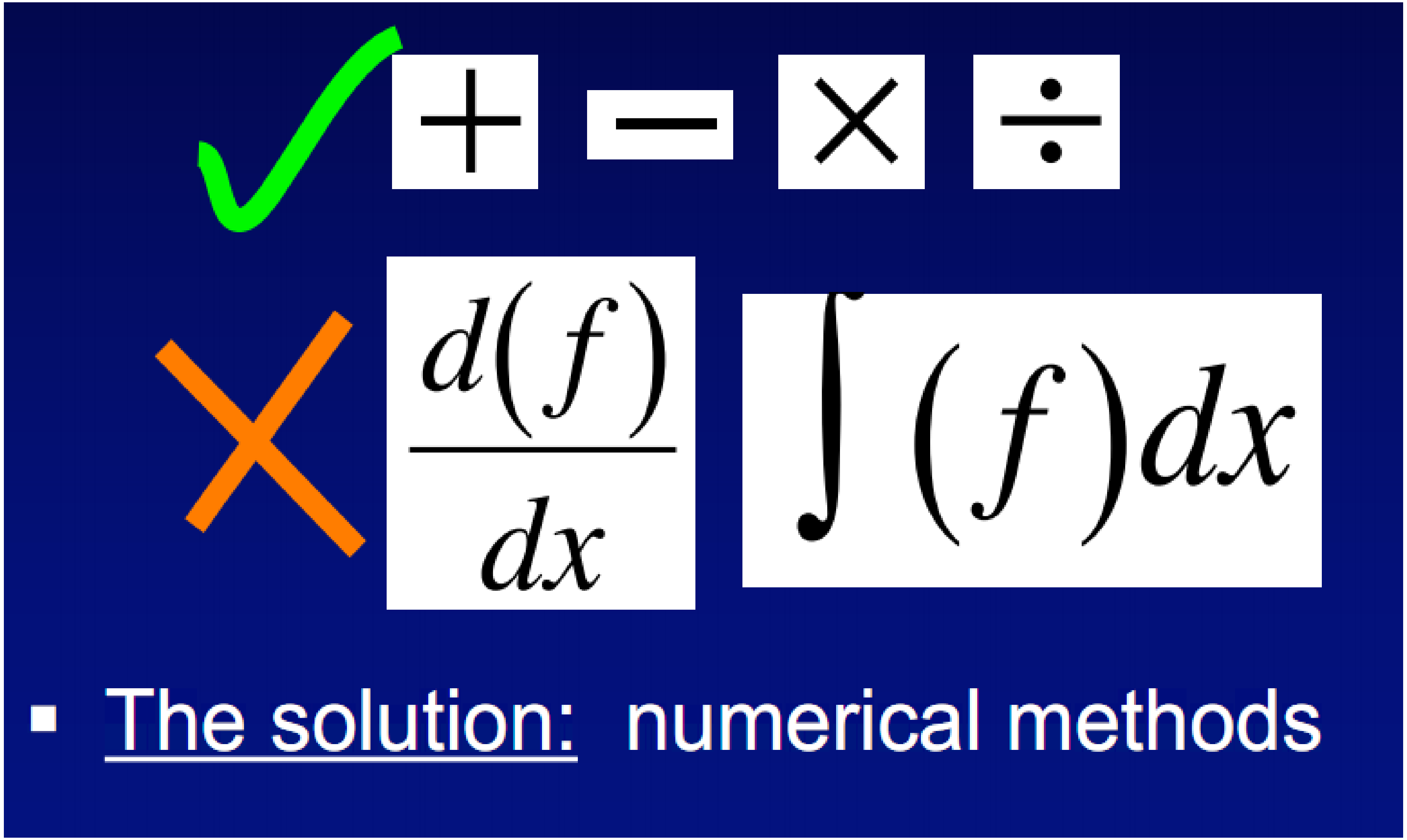
- *To Describe the Numerical Methods*
- *To Solve the set of PDEs through*
  - ✓ *Forward differencing*
  - ✓ *Backward differencing*
  - ✓ *Centered differencing*

Numerical Method

Numerical Methods

- Created because computers can perform arithmetic but not calculus
- Used to convert spatial and temporal derivative into equations that computers can solve
- Numerical methods directly affect model output, mostly at small scales

Cont….


$+ \quad - \quad \times \quad \div$
$\frac{d(f)}{dx}$
$\int (f)dx$
▪ The solution: numerical methods

- Numerical method is different from analytical because Analytical is exact but numerical is approximate.
- For example, some differential equations cannot be solved exactly (analytic or closed form solution) and we must rely on numerical techniques to solve them.
- Conceptually, a numerical method starts from an initial point and then takes a short step forward in time to find the next solution point.

- There are several distinct approaches to the formulation of computer methods for solving differential equations.
- We will confine ourselves to the finite difference method. Other approaches include finite element method and the spectral method.
- The central idea of the finite difference approach is to approximate the derivatives in the equation by differences between adjacent points in space or time, and thereby *reduce the differential equation to a difference equation.*

Finite Differences

Finite Differences

- The finite difference method is very widely used for numerically solving partial differential equations in science in general and in fluid mechanics in particular *because it is simple to implement.*

- In fact, it was the method Richardson (1922) used for the first attempt at weather forecasting by hand calculation.

- While methods using series expansion in terms of functions now compete with the finite difference method for the horizontal processing of meteorological fields, vertical discretization, to some extent, and time discretization still utilize finite differences.

Finite Differences

- In meteorology, the fundamental equations governing the atmospheric circulation appear, in general, to consist of sets of non-linear partial differential equations which do not have analytical solutions and are solved using numerical methods.

- The most common operations encountered during the solution of these equations are of the type of ***first and second derivative***, ***Jacobian*** and ***Laplacian***.

Forward, Backward And Centered Differences

$$f'(x) = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$f'(x) = \frac{f(x) - f(x-\Delta x)}{\Delta x}$$

- These are called the forward and backward differences

$$f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$$

Chapter 2: Numerical Methods

Objective

- *To Solve the Second order PDEs*
- *To write the Laplacian operators in Numerical format at 5-point stencil grid.*
- *Describe Jacobian operators*

Second order PDEs

The Laplacian Operator

The Laplacian Operator

- The Laplacian operator appears in many *diagnostic and prognostic equations in meteorology.*
- The application of its finite analog is found to be *very practical for the solution of many problems.*

The Jacobian operator

The Jacobian operator

- Jacobian is also a common operator encountered during the solution of many geophysical problems.
- It appears mostly in the non-linear advective terms.
- For example, the advection of vorticity by the horizontal wind in the vorticity equation

Let us consider a 5-point Jacobian $J(\psi,\zeta)$ given by:

$$J(\psi,\zeta)=\frac{\partial\psi}{\partial x}\frac{\partial\zeta}{\partial y}-\frac{\partial\psi}{\partial y}\frac{\partial\zeta}{\partial x}$$

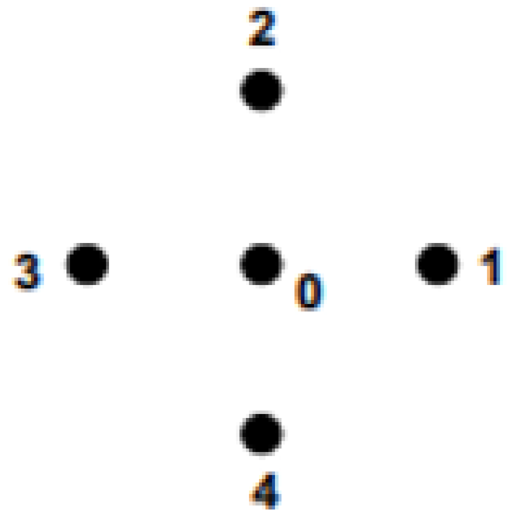


Fig. . A 5-Point Stencil for Jacobian

With the five grid points as shown in Fig. , the Jacobian $J(\psi,\zeta)$ can be expressed by

$$J(\psi,\zeta)\cong\frac{\psi_1-\psi_3}{2\Delta x}\frac{\zeta_2-\zeta_4}{2\Delta y}-\frac{\psi_2-\psi_4}{2\Delta y}\frac{\zeta_1-\zeta_3}{2\Delta x}$$

If $\Delta x=\Delta y=\Delta$, then

$$J(\psi,\zeta)=\frac{1}{4\Delta^2}[(\psi_1-\psi_3)(\zeta_2-\zeta_4)-(\psi_2-\psi_4)(\zeta_1-\zeta_3)]+\varepsilon(\Delta^2) \quad \ldots$$

….END

Thanks !

Chapter 2: Numerical Methods

Objective

- *To Be aware of Euler Scheme*
- *To Know Adams Bashforth Scheme*
- *To Use Leap Frog Scheme*
- *To Distinguish Implicit and Explicit Schemes*

Time Differencing Schemes

Time Differencing

The smooth flow of time implied by the left side of the equations of motion can be approximated by a sequence of discrete time steps, each of duration Δt. For example, the temperature tendency term on the left side of eq. (20.4) can be written as a centered time difference:

$$\frac{\Delta T}{\Delta t} \approx \frac{T(t+\Delta t) - T(t-\Delta t)}{2 \cdot \Delta t}$$

Time Differencing Schemes

- Atmospheric models generally require the solutions of partial differential equations, in which the dependent variables contain derivatives with respect to time.

- The basic concept of time integration is to predict the value of a time dependent function at time level (n+1) given it’s value at a time n

Euler's Method

Numerically approximate values for the solution of the initial-value problem $y' = F(x, y)$, $y(x_0) = y_0$, with step size h, at $x_n = x_{n-1} + h$, are

$$y_n = y_{n-1} + h \cdot F(x_{n-1}, y_{n-1})$$

Next Approximate Solution

Previous Approximate Solution ***Change in X*** ***Value of Differential equation at previous point***

Example Use Euler's method with step size $h = 0.2$ to find an approximation for $y(2)$, where y is a solution to the initial value problem

$$y' = x - y, \quad y(0) = 0.$$

i	x_i	$y_i = y_{i-1} + h(x_{i-1} - y_{i-1})$
0	0	0
1	0.2	0
2	0.4	0.04
3	0.6	0.1120
4	0.8	0.2096
5	1.0	0.327
6	1.2	0.4621
7	1.4	0.6097
8	1.6	0.7678
9	1.8	0.9342
10	2.0	1.107

Example 2

Consider the initial-value problem

$$y' = y - x, \quad y(0) = \tfrac{1}{2}.$$

Use Euler's method with (a) $h = 0.1$ and (b) $h = 0.05$ to obtain an approximation to

Solution: In this problem we have

$$f(x, y) = y - x, \qquad x_0 = 0, \qquad y_0 = \tfrac{1}{2}.$$

(a) Setting $h = 0.1$

$$y_{n+1} = y_n + 0.1(y_n - x_n).$$

Hence,

$$y_1 = y_0 + 0.1(y_0 - x_0) = 0.5 + 0.1(0.5 - 0) = 0.55,$$
$$y_2 = y_1 + 0.1(y_1 - x_1) = 0.55 + 0.1(0.55 - 0.1) = 0.595.$$

Continuing in this manner, we generate the approximations listed in Table where we have rounded the calculations to six decimal places.

n	x_n	y_n
1	0.1	0.55
2	0.2	0.595
3	0.3	0.6345
4	0.4	0.66795
5	0.5	0.694745
6	0.6	0.714219
7	0.7	0.725641
8	0.8	0.728205
9	0.9	0.721026
10	1.0	0.703129

b

n	x_n	y_n
2	0.1	0.54875
4	0.2	0.592247
6	0.3	0.629952
8	0.4	0.661272
10	0.5	0.685553
12	0.6	0.702072
14	0.7	0.710034
16	0.8	0.708563
18	0.9	0.696690
20	1.0	0.686525

Example 3

- Predict Y(1.3) in five decimal place

- Given $\frac{dy}{dt} = ty$ y(1) = 1 using $\Delta t = 0.1$

Ans = 1.36752

Adams Bash forth Scheme

- The Adams–Bashforth methods allow us explicitly to compute the approximate solution at an instant time from the solutions in previous instants.

$$y_{n+2} = y_{n+1} + \frac{3}{2}hf(t_{n+1}, y_{n+1}) - \frac{1}{2}hf(t_n, y_n).$$

- This method needs two values, Y(n+1) and Yn , to compute the next value, Yn+2.
- However, the initial value problem provides only one value, Y0 = 1.
- One possibility to resolve this issue is to use the Y1 computed by Euler's method as the second value.

$$y_{n+3} = y_{n+2} + h\left(\frac{23}{12}f(t_{n+2}, y_{n+2}) - \frac{16}{12}f(t_{n+1}, y_{n+1}) + \frac{5}{12}f(t_n, y_n)\right),$$

$$y_{n+4} = y_{n+3} + h\left(\frac{55}{24}f(t_{n+3}, y_{n+3}) - \frac{59}{24}f(t_{n+2}, y_{n+2}) + \frac{37}{24}f(t_{n+1}, y_{n+1}) - \frac{9}{24}f(t_n, y_n)\right),$$

$$y_{n+5} = y_{n+4} + h\left(\frac{1901}{720}f(t_{n+4}, y_{n+4}) - \frac{2774}{720}f(t_{n+3}, y_{n+3}) + \frac{2616}{720}f(t_{n+2}, y_{n+2}) - \frac{1274}{720}f(t_{n+1}, y_{n+1}) + \frac{251}{720}f(t_n, y_n)\right).$$

Leapfrog scheme (Central time central space)

$$\frac{\partial u}{\partial x}$$

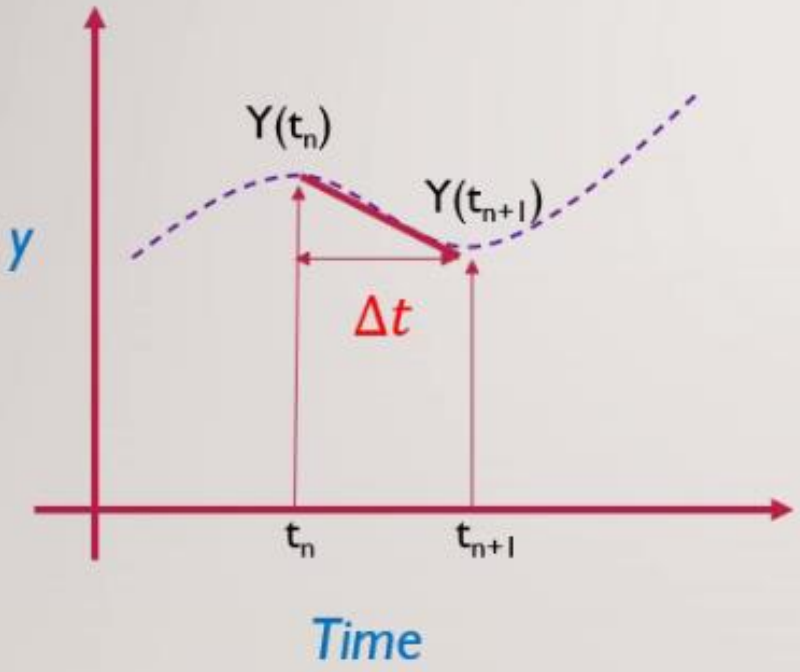
$\frac{\partial u}{\partial x}+$ a) $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$ *Forward, backward and centered finite difference method*

Explicit and Implicit Schemes

Explicit and Implicit

- A finite difference scheme is said to be *explicit* when it can be computed forward in time using quantities from previous time steps
- In *implicit* finite-difference schemes, the output of the time-update (above) depends on itself, so a causal recursive computation is not specified

IMPLICIT VS EXPLICIT SCHEME



Explicit :-

Solution at next step is found using Slope at Known Time step

$$\underbrace{y_{n+1}}_{\text{Unknown}} = \underbrace{y_n}_{\text{Known}} + \Delta t.\underbrace{y_n'}_{\text{Known}}$$

Implicit :-

Solution at next step is found using Slope at unknown Time step

$$\underbrace{y_{n+1}}_{\text{Unknown}} = \underbrace{y_n}_{\text{Known}} + \Delta t.\underbrace{y_{n+1}'}_{\text{Unknown}}$$

Methods' like newton Raphson are used for iteration to find unknowns

Explicit

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

In explicit finite difference schemes, the temperature at time n+1 depends explicitly on the temperature at time n. The explicit finite difference discretization of the above equation :

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \kappa \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}$$

$$T_i^{n+1} = T_i^n + \kappa \Delta t \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}$$

The major advantage of explicit finite difference methods is that they are relatively simple and computationally fast. However, the main drawback is that stable solutions are obtained only when

$$0 < \frac{\kappa \Delta t}{\Delta x^2} < 0.5$$

Implicit

- In implicit nite dierence schemes, the spatial derivatives $\frac{\partial^2 T}{\partial x^2}$ are evaluated (at least partially) at the new timestep. The simplest implicit discretization of the above equation is :

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \kappa \frac{T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}}{\Delta x^2}$$

Example

- Approximate the value of Y(0.2) given problem $\frac{dy}{dx} = x + y$ Y(0) = 1

Take h= 0.1 (correct to three decimal place) use Implicit Euler's method

Ans = 1.269

Stability of Numerical Methods

- If a numerical method is **stable**

✓ A small changes in the initial condition result in small changes in the computed solution.

- The numerical method is **unstable**

✓ If a small changes in the initial condition result in large changes in the computed solution.

Stability

- Whenever a numerical method is applied to solve a differential equation, one needs to examine the stability of the numerical solution.
- This type of *numerical or computational stability* problem can be investigated by obtaining the *amplification factor(λ)* , the magnitude of which would determine whether a numerical scheme is stable or not. The general stability criteria are:

$$\begin{Bmatrix} \text{Unstable scheme} & \text{if } |\lambda| > 1, \\ \text{Neutral scheme} & \text{if } |\lambda| = 1, \\ \text{Stable scheme} & \text{if } |\lambda| < 1. \end{Bmatrix}$$

Example

- $y' = \lambda y$, with $y(t_0) = y_0$:

$$y_{n+1} = y_n + \Delta t f(t_n, y_n)$$
$$y_{n+1} = y_n + \Delta t(\lambda y_n)$$
$$y_{n+1} = (1 + \lambda \Delta t) y_n$$

- The solution is stable when $|1 + \lambda \Delta t| \leq 1$

….END

Thanks !